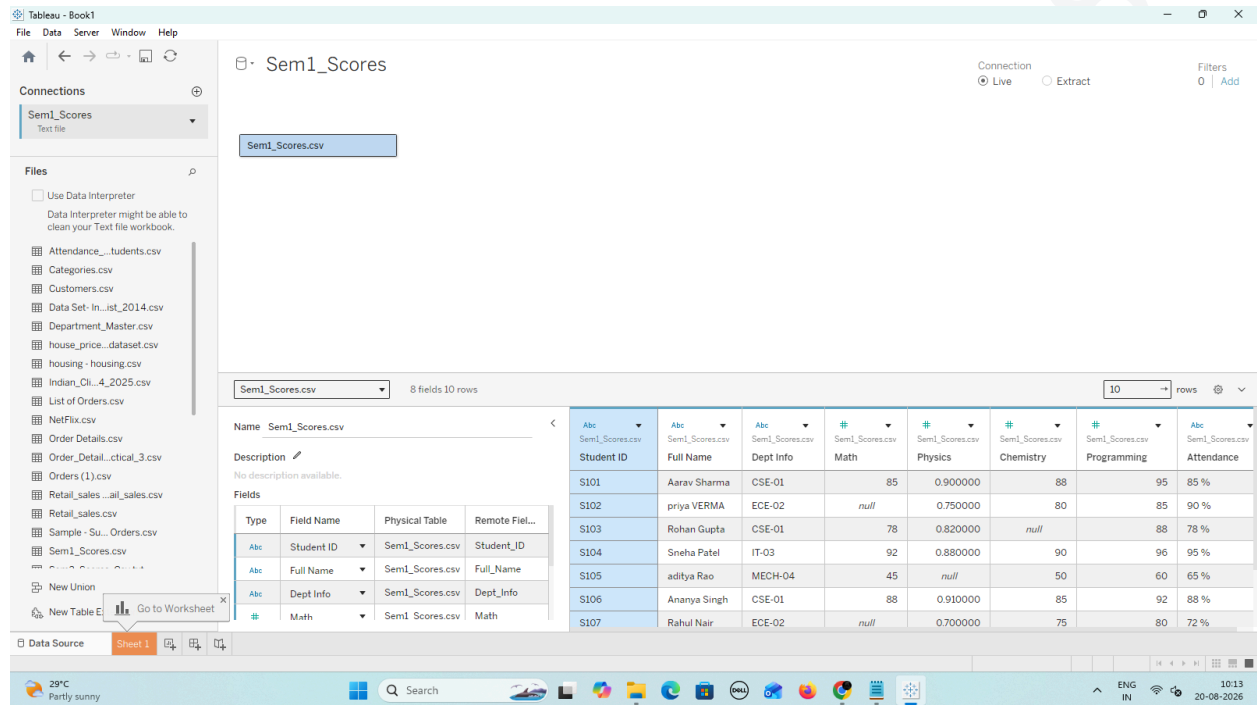


SHETH L.U.J & SIR M.V

AIM :- Cleaning and Structuring Data for Analysis

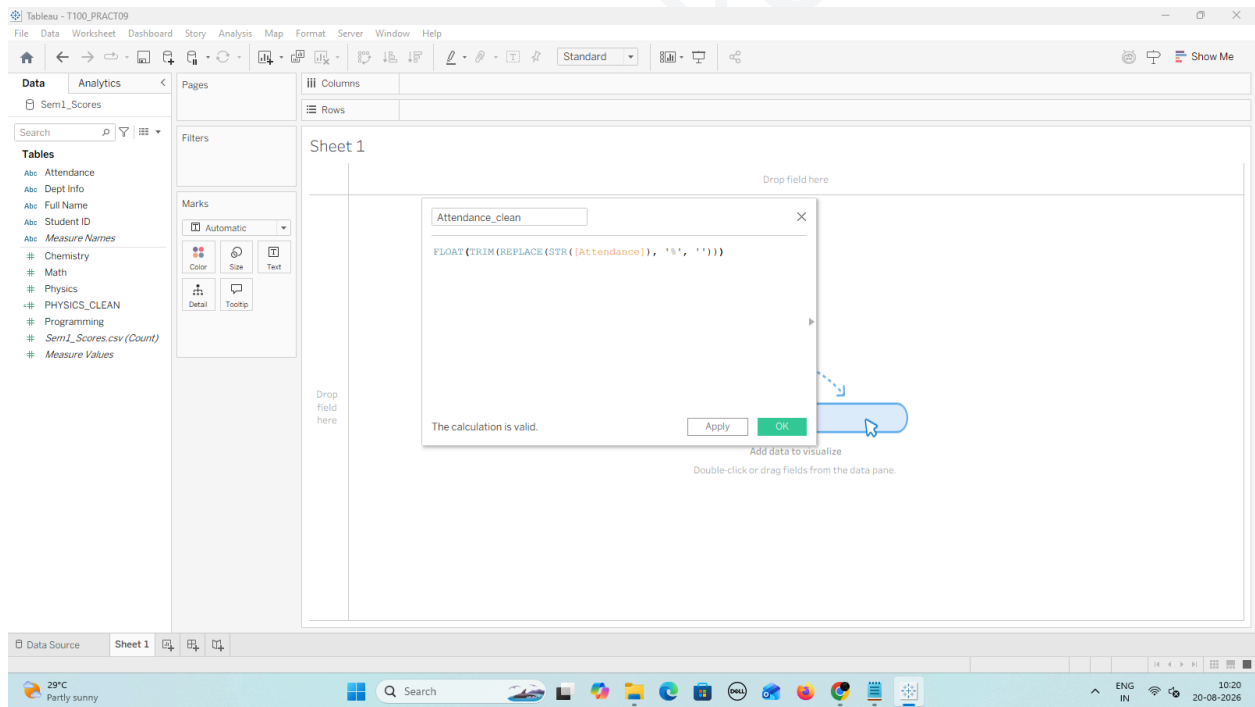
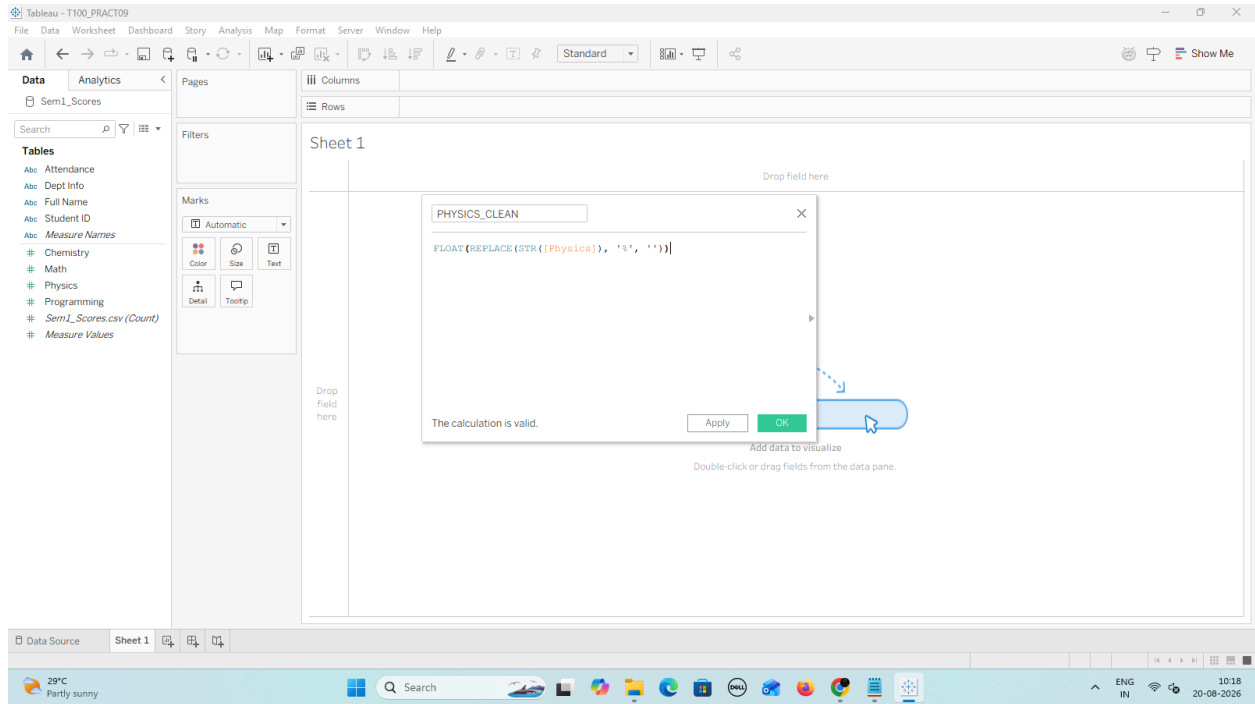
- Identify and handle missing or inconsistent data
- Convert wide data into tall format
- Perform union of multiple files
- Execute cross-database joins
- Restructure data for better visualization
- Handle different levels of detail in datasets
- Apply advanced data cleaning techniques



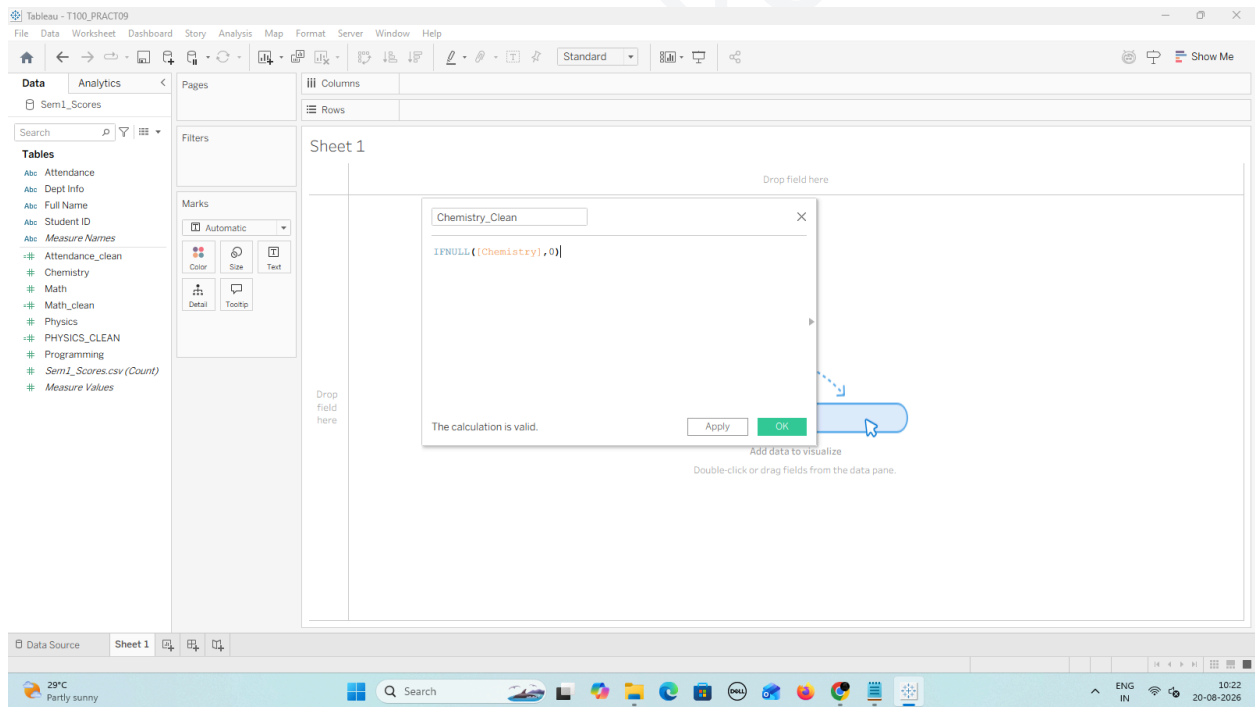
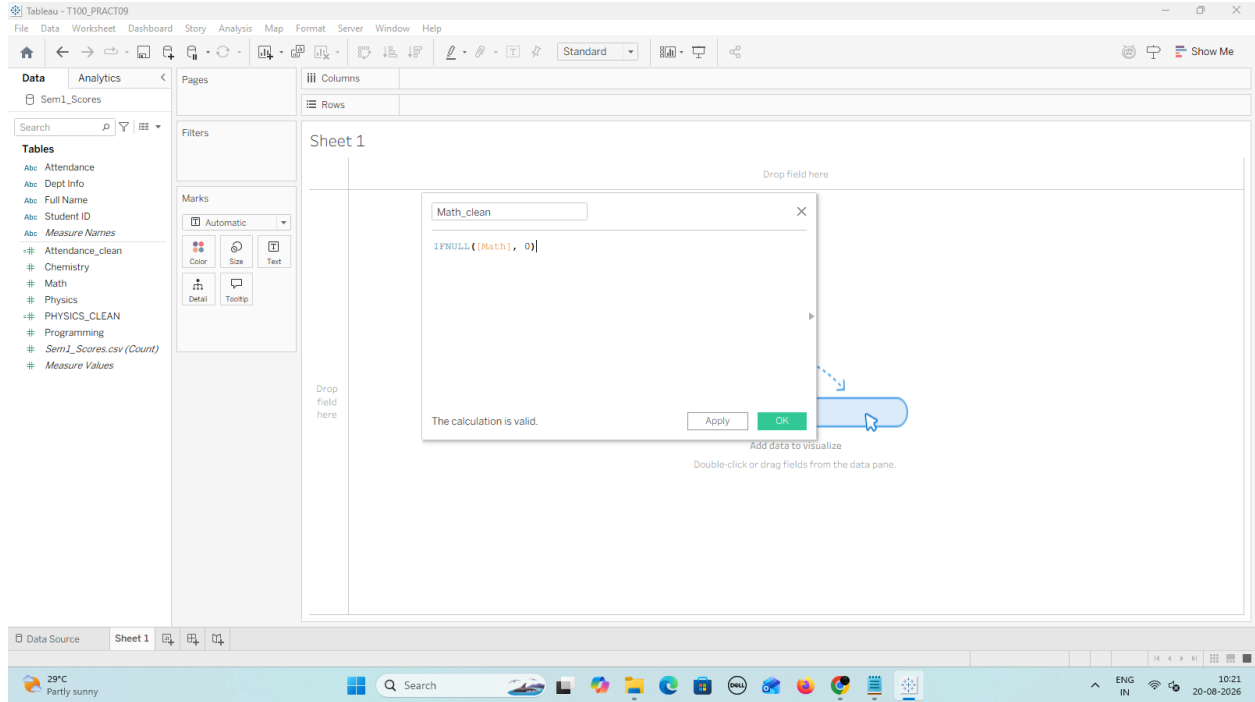
The screenshot shows the Tableau Desktop interface. On the left, the 'Connections' pane lists 'Sem1_Scores' as a Text File. Below it, the 'Files' pane shows a list of CSV files. The main view displays the 'Sem1_Scores' data source with 8 fields and 10 rows. The data is presented in a table with columns for Student ID, Full Name, Dept Info, Math, Physics, Chemistry, Programming, and Attendance. The data is as follows:

Student ID	Full Name	Dept Info	Math	Physics	Chemistry	Programming	Attendance
S101	Aarav Sharma	CSE-01	85	0.900000	88	95	85 %
S102	priya VERMA	ECE-02	null	0.750000	80	85	90 %
S103	Rohan Gupta	CSE-01	78	0.820000	null	88	78 %
S104	Sneha Patel	IT-03	92	0.880000	90	96	95 %
S105	aditya Rao	MECH-04	45	null	50	60	65 %
S106	Ananya Singh	CSE-01	88	0.910000	85	92	88 %
S107	Rahul Nair	ECE-02	null	0.700000	75	80	72 %

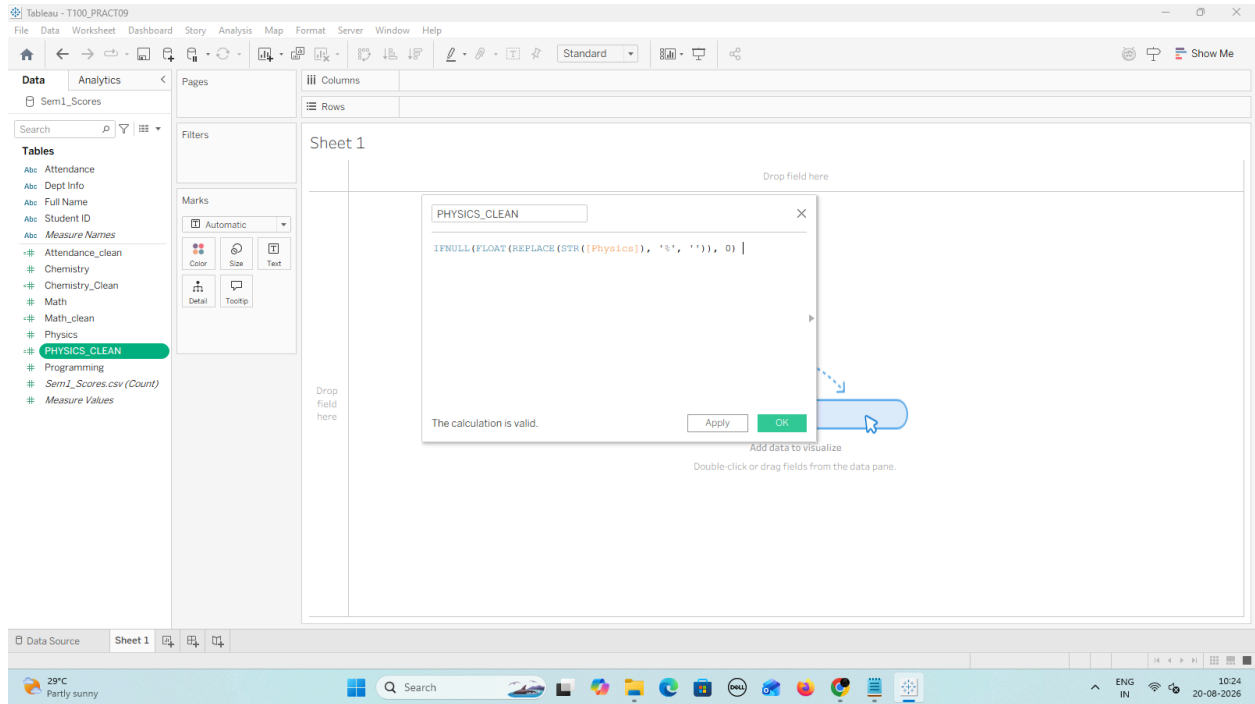
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SHETH L.U.J & SIR M.V



SHETH L.U.J & SIR M.V



The screenshot shows the Tableau Desktop interface with the 'Sem1_Scores' data source selected. The data table is displayed with 10 rows and 12 fields. The table includes columns for Student ID, Full Name, Dept Info, Math, Physics, Chemistry, Programming, and Attendance.

Student ID	Full Name	Dept Info	Math	Physics	Chemistry	Programming	Attendance
S101	Aarav Sharma	CSE-01	85	0.900000	88	95	85 %
S102	priya VERMA	ECE-02	null	0.750000	80	85	90 %
S103	Rohan Gupta	CSE-01	78	0.820000	null	88	78 %
S104	Sneha Patel	IT-03	92	0.880000	90	96	95 %
S105	aditya Rao	MECH-04	45	null	50	60	65 %
S106	Ananya Singh	CSE-01	88	0.910000	85	92	88 %
S107	Rahul Nair	ECE-02	null	0.700000	75	80	72 %

TABLOU - TIOO_PRACT09

File Data Server Window Help

Sem1_Scores

Connections
Sem1_Scores
Text file

Files
☐ Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

- Attendance_students.csv
- Categories.csv
- Customers.csv
- Data Set-In Ist_2014.csv
- Department_Master.csv
- house_price_dataset.csv
- housing-housing.csv
- Indian_Cli_4_2025.csv
- List of Orders.csv
- NetFlix.csv
- Order Details.csv
- Order_Detail_ctical_3.csv
- Orders (1).csv
- Retail_sales_all_sales.csv
- Retail_sales.csv
- Sample-Sales.Orders.csv
- Sem1_Scores.csv
- New Union
- New Table Extension

Connection
☒ Live ☐ Extract
Filters
0 | Add

Sem1_Scores.csv

Relate tables
Drag tables near Sem1_Scores.csv to relate them

Add a base table
Drag out additional base tables for multi-fact analysis.

Need help adding data? [Learn more](#)

Sem1_Scores.csv 12 fields 10 rows

Name	Sem1_Scores.csv	Description
Akt	Dept Info	Sem1_Scores.csv Dept Info
#	Math	Sem1_Scores.csv Math
#	Physics	Sem1_Scores.csv Physics
#	Chemistry	Sem1_Scores.csv Chemistry
#	Programming	Sem1_Scores.csv Programming

Student ID	Full Name	Dept Info	Math	Physics	Chemistry	Programming	Attendance	CALCULATION PHYSICS_CLEAN
S101	Aarav Sharma	CSE-01	85	0.900000	88	95	85 %	0.900000
S102	priya VERMA	ECE-02	null	0.750000	80	85	90 %	0.750000
S103	Rohan Gupta	CSE-01	78	0.820000	null	88	78 %	0.820000
S104	Sneha Patel	IT-03	92	0.880000	90	96	95 %	0.880000
S105	aditya Rao	MECH-04	45	null	50	60	65 %	0.000000
S106	Ananya Singh	CSE-01	88	0.910000	85	92	88 %	0.910000
S107	Rahul Nair	ECE-02	null	0.700000	75	80	72 %	0.700000
S108	Kavya Reddy	IT-03	95	0.940000	92	98	98 %	0.940000

Data Source Sheet 1

Purvi .R. Karkera T086

SHETH L.U.J & SIR M.V

Tableau - T100_PRACT09

File Data Server Window Help

Connections

Sem1_Scores
Text file

Files

☐ Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Sem1_Scores.csv
Sem2_Scores.Csv.txt
Sem2_Scores.csv
Sem2_Scores.csv
startup_funding.csv
store_sales_data (2).csv
StressLevel...set (1).csv
StressLevel_Dataset.csv
Student_Activities.csv
Student_Performance.csv
Students.csv
supermarket_sales (2).csv
supermarket_sales.csv
table-1785...716728.csv
table-1785...987032.csv
Titanic-Dataset.csv
Titanic.csv
weatherHistory.csv

New Union
New Table Extension

Sem1_Scores

Relate tables
Drag tables near Sem1_Scores.csv to relate them

Add a base table
Drag out additional base tables for multi-fact analysis.

Need help adding data? [Learn more](#)

Sem1_Scores.csv 10 fields 40 rows

Name	Sem1_Scores.csv
Student ID	Aarav Sharma
Full Name	CSE-01
Dept Info	85 %
Attendance	Chemistry
Pivot	Pivot Field Names
Pivot Field Values	88.0000
PHYSICS_CLEAN	Calculation
Attendance_clean	Calculation

40 rows

30°C Partly sunny

Q Search

ENG IN 10:53 20-08-2026

Tableau - T100_PRACT09

File Data Server Window Help

Connections

Sem1_Scores
Text file

Files

☐ Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Sem1_Scores.csv
Sem2_Scores.Csv.txt
Sem2_Scores.csv
Sem2_Scores.csv
startup_funding.csv
store_sales_data (2).csv
StressLevel...set (1).csv
StressLevel_Dataset.csv
Student_Activities.csv
Student_Performance.csv
Students.csv
supermarket_sales (2).csv
supermarket_sales.csv
table-1785...716728.csv
table-1785...987032.csv
Titanic-Dataset.csv
Titanic.csv
weatherHistory.csv

New Union
New Table Extension

Sem1_Scores

Relate tables
Drag tables near Sem1_Scores.csv to relate them

Add a base table
Drag out additional base tables for multi-fact analysis.

Need help adding data? [Learn more](#)

Sem1_Scores.csv 10 fields 40 rows

Name	Sem1_Scores.csv
Student ID	Aarav Sharma
Full Name	CSE-01
Dept Info	85 %
Attendance	Chemistry
Pivot	Pivot subject
Pivot Score	88.0000
PHYSICS_CLEAN	Calculation
Attendance_clean	Calculation

40 rows

30°C Partly sunny

Q Search

ENG IN 10:55 20-08-2026

Tableau - T100_PRACT9

File Data Server Window Help

Home < > < > < >

Connections

Sem1_Scores
Text file

Files

☐ Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

- Sem1_Scores.csv
- Sem2_Scores.Csv.txt
- Sem2_Scores.csv
- startup_funding.csv
- store_sales_data (2).csv
- StressLevel_set (1).csv
- StressLevel_Dataset.csv
- Student_Activities.csv
- Student_Performance.csv
- Students.csv
- supermarket_sales (2).csv
- supermarket_sales.csv
- table-1785...716728.csv
- table-1785...987032.csv
- Titanic-Dataset.csv
- Titanic.csv
- weatherHistory.csv

+ New Union

+ New Table Extension

Sem1_Scores

Relate tables
Drag tables near Sem1_Scores.csv to relate them

Add a base table
Drag out additional base tables for multi-fact analysis.

Need help adding data? [Learn more](#)

Number (decimal)
Number (whole)
Date & Time
Date
✓ String
Spatial
Boolean

Fields

✓ Default

Geographic Role
Image Role

Physical Table
Remote...

Score
PHYSICS_CLEAN
Attendance_clean

Pivot
Calculation
Calculation

Student ID	Full Name	Deg1 Info	Attendance	Subject	Post Score	PHYSICS_CLEAN	Attendance_clean
S101	Aarav Sharma	CSE-01	85 %	Chemistry	88.0000	null	85.00
S101	Aarav Sharma	CSE-01	85 %	Math	85.0000	null	85.00
S101	Aarav Sharma	CSE-01	85 %	Physics	9.0000	null	85.00
S101	Aarav Sharma	CSE-01	85 %	Programming	95.0000	null	85.00
S102	priya VERMA	ECE-02	90 %	Chemistry	80.0000	null	90.00
S102	priya VERMA	ECE-02	90 %	Math	null	null	90.00
S102	priya VERMA	ECE-02	90 %	Physics	0.7500	null	90.00
S102	priya VERMA	ECE-02	90 %	Programming	85.0000	null	90.00

Data Source Sheet 1

30°C Partly sunny

Search

10:56 20-08-2024

Purvi .R. Karkera T086

SHETH L.U.J & SIR M.V

- c) Perform union of multiple files
- d) Execute cross-database joins

The screenshot shows the Tableau Desktop interface with the 'Sem1_Scores' data source selected. The main workspace displays a table with 12 fields and 10 rows. The table structure is as follows:

Name	Sem1_Scores.csv	Student ID	Full Name	Dept Info	Math	Physics	Chemistry	Programming	Attendance	PHYSICS_CLEAN
S103	Rohan Gupta	CSE-01	78	0.820000	null	88	78 %	0.82000		
S104	Sneha Patel	IT-03	92	0.880000	90	96	95 %	0.88000		
S105	aditya Rao	MECH-04	45	null	50	60	65 %	0.00000		
S106	Ananya Singh	CSE-01	88	0.910000	85	92	88 %	0.91000		
S107	Rahul Nair	ECE-02	null	0.700000	75	80	72 %	0.70000		
S108	Kavya Reddy	IT-03	95	0.940000	92	98	98 %	0.94000		
S109	Vikram Joshi	CSE-01	67	0.600000	58	72	80 %	0.60000		
S110	Pooja Iyer	ECE-02	82	0.850000	80	89	91 %	0.85000		

The screenshot shows the 'Union' dialog box in Tableau Desktop. The 'Specific (manual)' tab is selected, and two data sources are listed: 'Sem1_Scores.csv' and 'Sem2_Scores.csv'. The 'Wildcard (automatic)' tab is also visible. The dialog box indicates that there are 2 tables in the union.

SHETH L.U.J & SIR M.V

Tableau - T100_PRACT09

File Data Server Window Help

Connections

Sem1_Scores
Text file

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Sem1_Scores.csv
Sem2_Scores.Csv.txt
Sem2_Scores.csv
startup_funding.csv
store_sales_data (2).csv
StressLevel...set (1).csv
StressLevel...Dataset.csv
Student_Activities.csv
Student_Performance.csv
Students.csv
supermarket_sales (2).csv
supermarket_sales.csv
table-1785...716728.csv
table-1785...987032.csv
Titanic-Dataset.csv
Titanic.csv
weatherHistory.csv

New Union
New Table Extension

Sem1_Scores

Connection
Live Extract
Filters
0 Add

Union

Relate tables
Drag tables near Union to relate them

Add a base table
Drag out additional base tables for multi-fact analysis.

Need help adding data? Learn more

Union 13 fields 0 rows

Name Union

Description
No description available.

Fields

Type	Field Name	Physical Table	Remote FL...
Abc	Student_ID	Union	Student_ID
Abc	Full_Name	Union	Full_Name
Abc	Dept_Info	Union	Dept_Info
#	Math	Union	Math
#	Physics	Union	Physics

Abc Union Student_ID
Abc Union Full_Name
Abc Union Dept_Info
Union Math
Union Physics
Union Chemistry
Union Programming
Abc Union Attendance
Abc Union Table Name
Calcul PHYS

29°C Partly sunny

Search

ENG IN 10:41 20-08-2026

Tableau - T100_PRACT09

File Data Server Window Help

Connections

Sem1_Scores
Text file

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Sem1_Scores.csv
Sem2_Scores.Csv.txt
Sem2_Scores.csv
startup_funding.csv
store_sales_data (2).csv
StressLevel...set (1).csv
StressLevel...Dataset.csv
Student_Activities.csv
Student_Performance.csv
Students.csv
supermarket_sales (2).csv
supermarket_sales.csv
table-1785...716728.csv
table-1785...987032.csv
Titanic-Dataset.csv
Titanic.csv
weatherHistory.csv

New Union
New Table Extension

Sem1_Scores

Connection
Live Extract
Filters
0 Add

Union

Relate tables
Drag tables near Union to relate them

Add a base table
Drag out additional base tables for multi-fact analysis.

Need help adding data? Learn more

Union 13 fields 0 rows

Name Union

Description
No description available.

Fields

Type	Field Name	Physical Table	Remote FL...
Abc	Student_ID	Union	Student_ID
Abc	Full_Name	Union	Full_Name
Abc	Dept_Info	Union	Dept_Info
#	Math	Union	Math
#	Physics	Union	Physics

Abc Union Student_ID
Abc Union Full_Name
Abc Union Dept_Info
Union Math
Union Physics
Union Chemistry
Union Programming
Abc Union Attendance
Abc Union Table Name
Calcul PHYS

29°C Partly sunny

Search

ENG IN 10:42 20-08-2026

Sem1_Scores

Connections
Sem1_Scores
Text file

Files

- Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.
- housing - housing.csv
- Indian_Cli_4_2025.csv
- List of Orders.csv
- NetFix.csv
- Order Details.csv
- Order_Detail_ctical_3.csv
- Orders (1).csv
- Retail_sales_all_sales.csv
- Retail_sales.csv
- Sample - Su... Orders.csv
- Sem1_Scores.csv
- Sem2_Scores.Csv.txt
- Sem2_Scores.csv
- startup_funding.csv
- store_sales_data (2).csv
- StressLevel_set (1).csv
- StressLevel_Dataset.csv
- New Union
- New Table Extension

Union

Relate tables
Drag tables near Union to relate them

Add a base table
Drag out additional base tables for multi-fact analysis.

Need help adding data? [Learn more](#)

Union 14 fields 20 rows

Name	Union	Student_ID	Full_Name	Dept_Info	Math	Physics	Chemistry	Programming	Attendance	Table Name
S108	Kavya Reddy	IT-03	95	0.940000	92	98	98 %	Sem1_Scores.csv		
S109	Vikram Joshi	CSE-01	67	0.600000	58	72	80 %	Sem1_Scores.csv		
S110	Pooja Iyer	ECE-02	82	0.850000	80	89	91 %	Sem1_Scores.csv		
S101	Aarav Sharma	CSE-01	88	0.920000	90	98	88 %	Sem2_Scores.c..		
S102	Priya Verma	ECE-02	79	0.780000	82	88	92 %	Sem2_Scores.c..		
S103	Rohan Gupta	CSE-01	82	0.850000	80	90	80 %	Sem2_Scores.c..		
S104	Sneha Patel	IT-03	95	0.900000	94	98	96 %	Sem2_Scores.c..		
S105	Aditya Rao	MFCN-04	63	0.590000	65	66	70 %	Sem2_Scores.c..		

Purvi .R. Karkera T086

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Tableau - T100_PRACT09

File Data Server Window Help

Sem1_Scores

Connections
Sem1_Scores
Text file

Files
☐ Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

- Attendance_students.csv
- Categories.csv
- Customers.csv
- Data Set - In...st_2014.csv
- Department_Master.csv
- house_price_dataset.csv
- housing_housing.csv
- Indian_Cli...4_2025.csv
- List of Orders.csv
- Netflix.csv
- Order Details.csv
- Order_Detail...ctical_3.csv
- Orders (1).csv
- Retail_sales_all_sales.csv
- Retail_sales.csv
- Sample - Su...Orders.csv
- Sem1_Scores.csv
- New Union
- New Table Extension

Connection
☒ Live ☐ Extract Filters
0 | Add

Union Department_Master.csv

Union — Departme... 20 rows

Name	Union	Operator	Department_Master...	Abs Student_ID	Abs Full_Name	Abs Dept_Info	Math	Physics	Chemistry	Programming	Abs Attendance	Abs Table Name	Cali PH)
Union		=	Dept Code	S101	Aarav Sharma	CSE-01	85	0.900000	88	95	85 %	Sem1_Scores.csv	
				S102	priya VERMA	ECE-02	null	0.750000	80	85	90 %	Sem1_Scores.csv	
				S103	Rohan Gupta	CSE-01	78	0.820000	null	88	78 %	Sem1_Scores.csv	
				S104	Sneha Patel	IT-03	92	0.880000	90	96	95 %	Sem1_Scores.csv	
				S105	aditya Rao	MECH-04	45	null	50	60	65 %	Sem1_Scores.csv	
				S106	Ananya Singh	CSE-01	88	0.910000	85	92	88 %	Sem1_Scores.csv	
				S107	Rahul Nair	ECE-02	null	0.700000	75	80	72 %	Sem1_Scores.csv	
				S108	Kavya Reddy	IT-03	95	0.940000	92	98	98 %	Sem1_Scores.csv	

> Performance Options
Remove Relationship

Data Source Sheet 1

T100 - T100_PRACT09

File

Data

Server

Window

Help

Home

Back

Forward

Refresh

Sem1_Scores

Text file

Connections

Sem1_Scores

Text file

Files

Use Data Interpreter

Data Interpreter might be able to clean your Text file workbook.

Netflix.csv

Order_Details.csv

Order_Detail...ctical_3.csv

Orders (1).csv

Retail_sales ...all_sales.csv

Retail_sales.csv

Sample · Su...Orders.csv

Sem1_Scores.csv

Sem2_Scores.csv.txt

Sem2_Scores.csv

startup_funding.csv

store_sales_data (2).csv

StressLevel...set (1).csv

StressLevel_Dataset.csv

Student_Activities.csv

Student_Performance.csv

Students.csv

New Union

New Table Extension

Sem1_Scores

Department_Master.csv

Student_Activities.csv

Union

Student_A...

10

rows

Name

Student_Activities.csv

How do relationships differ from joins? Learn more

Union

Operator

Student_Activities...

Abc Student_ID

=

Abc Student ID (Stu...

Remove Relationship

Abc Student_Activities.csv

Student ID (Student Activi

Abc Student_Activities.csv

Club Membership

Student_Activities.csv

Projects Completed

Student_Activities.csv

Certifications

S101	Coding Club	3	2
S102	Robotics Club	2	1
S103	Coding Club	2	1
S104	Debate Society	4	3
S105	Sports Club	1	0
S106	Coding Club	3	2
S107	Robotics Club	2	1
S108	Debate Society	4	3

Data Source

Sheet 1

Grid

Table

30°C

Partly sunny

Windows

Search

Taskbar

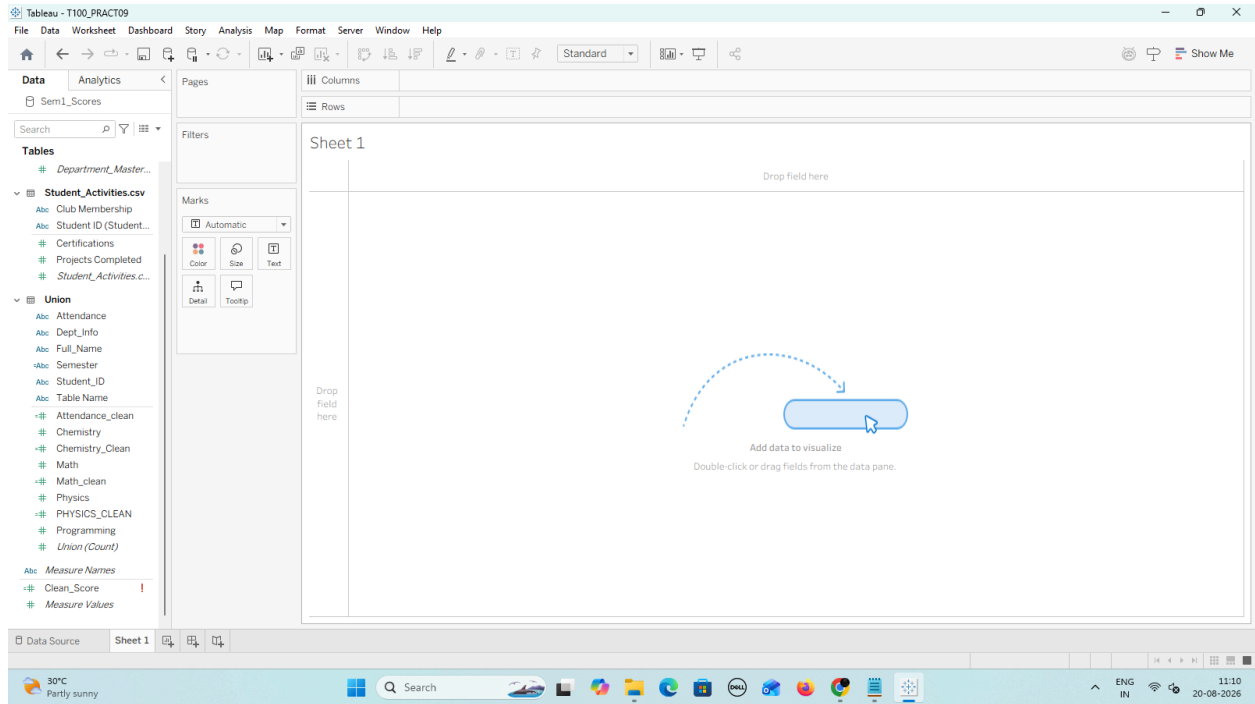
ENG

IN

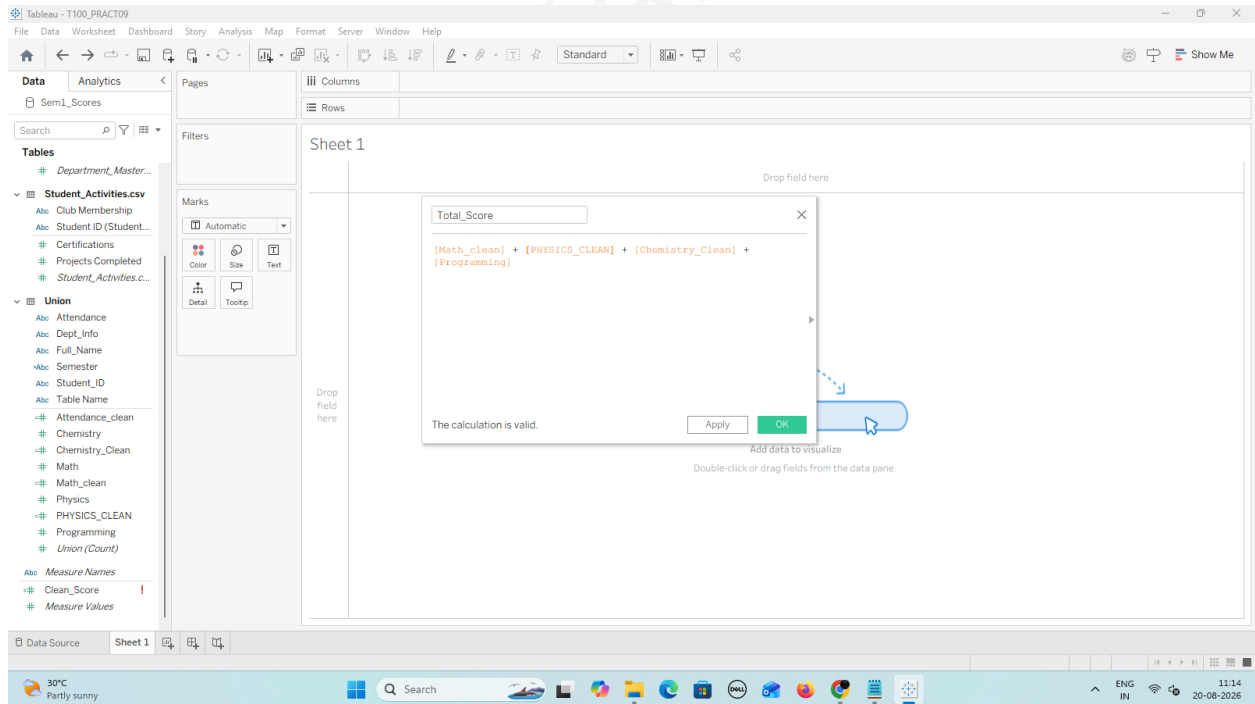
11:07

20-08-2028

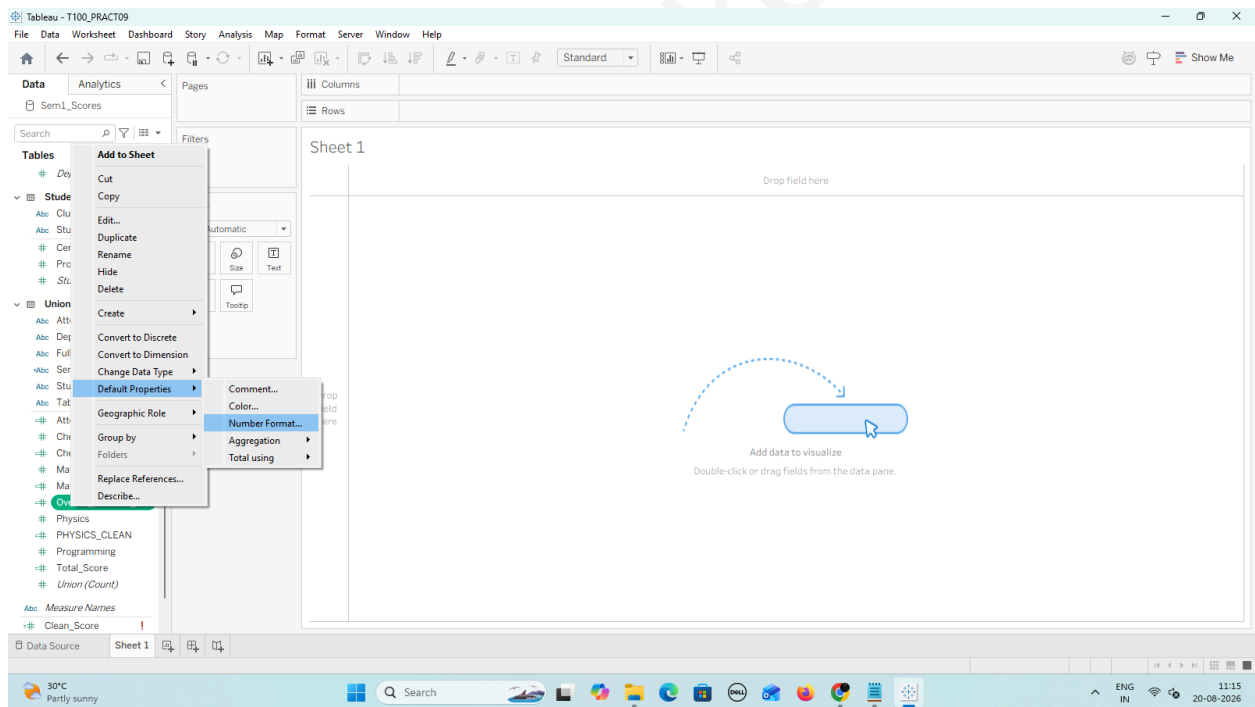
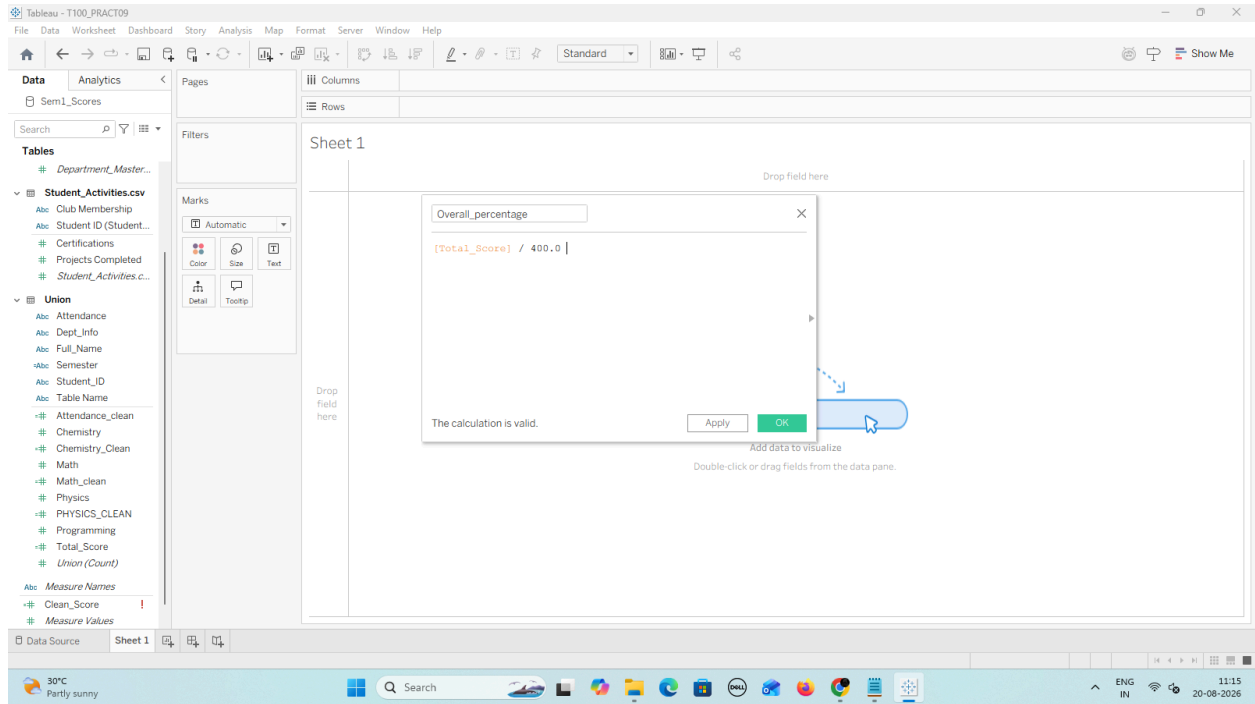
SHETH L.U.J & SIR M.V



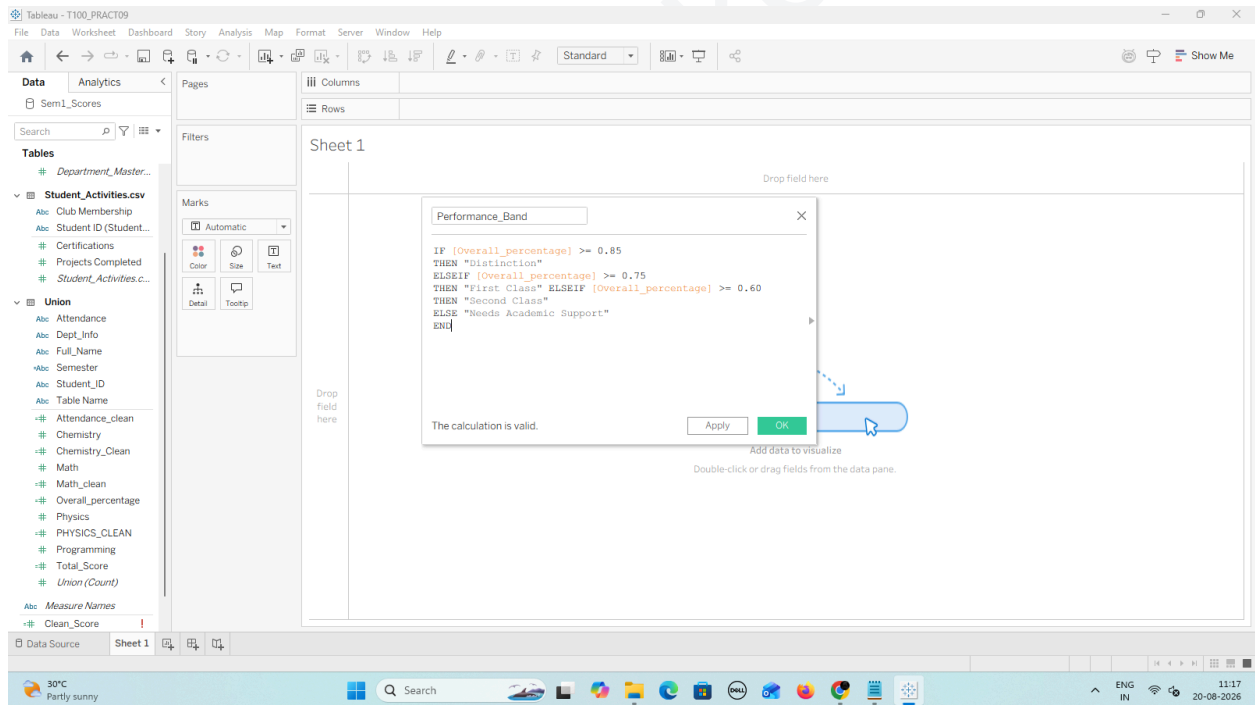
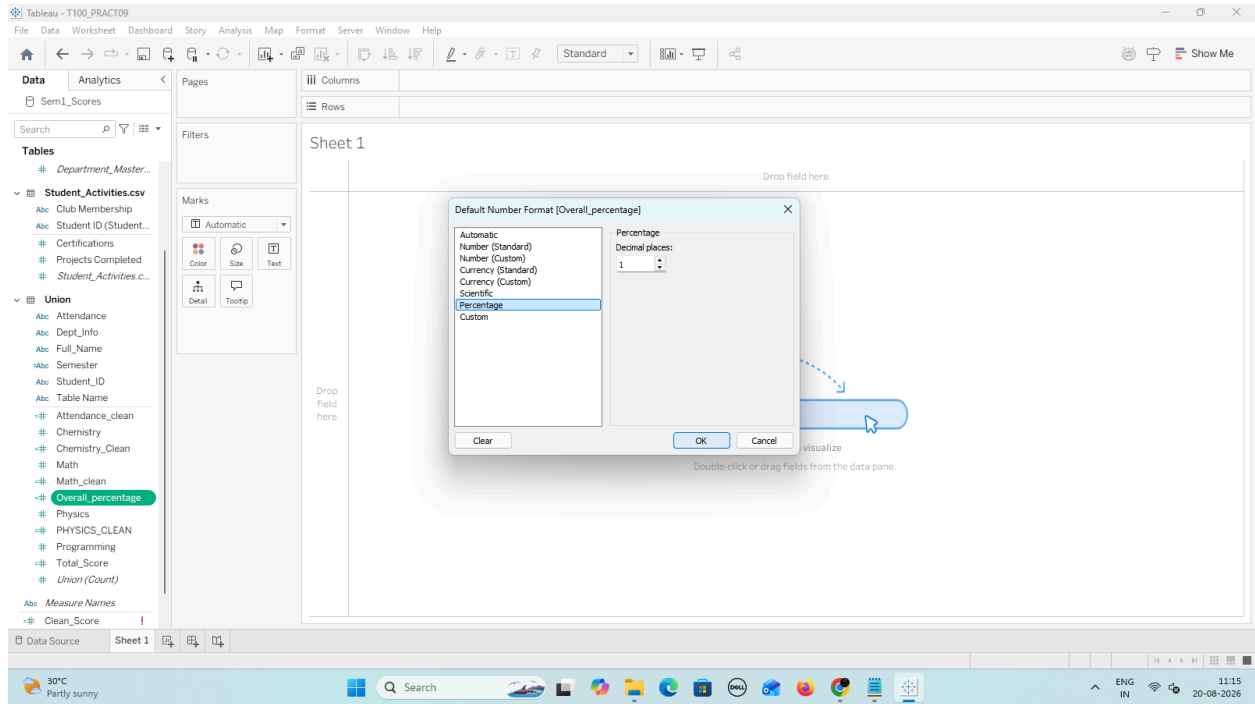
- e) Restructure data for better visualization
- f) Handle different levels of detail in datasets



SHETH L.U.J & SIR M.V



SHETH L.U.J & SIR M.V



SHETH L.U.J & SIR M.V

The screenshot shows the Tableau Desktop interface with the 'Attendance_Status' calculation dialog box open. The dialog box contains the following code:

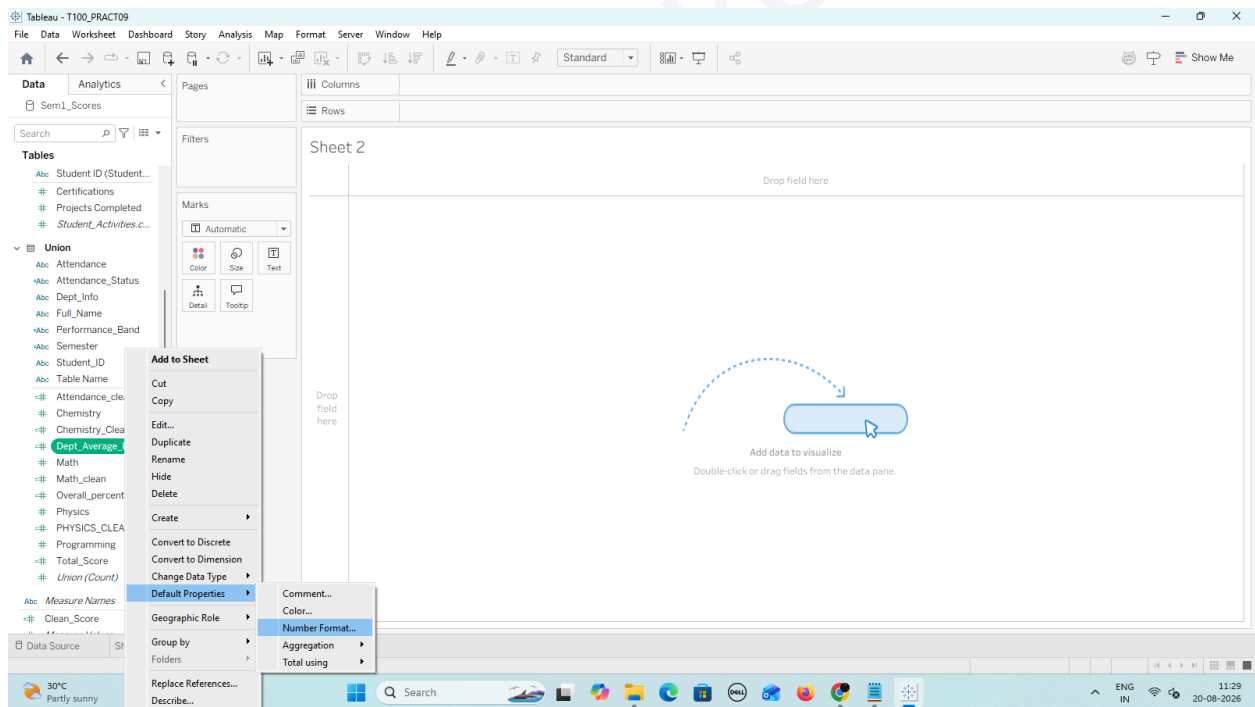
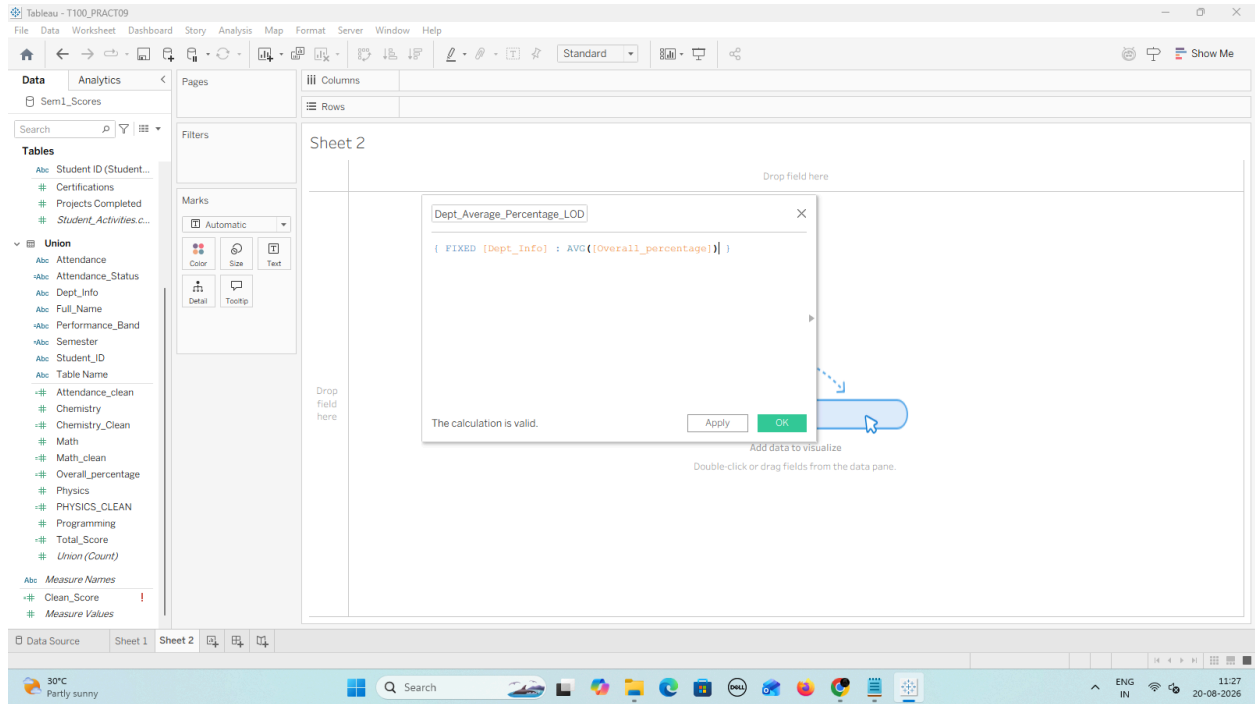
```
IF [Attendance_clean] >= 75
THEN "Eligible (Good Standing)"
ELSE "Shortage Warning (<75%)"
END
```

The dialog box also includes an 'Apply' button, an 'OK' button, and a 'Cancel' button. The background shows the Tableau interface with the 'Student_Activities.csv' data source selected in the 'Data' pane. The 'Marks' card is set to 'Automatic'.

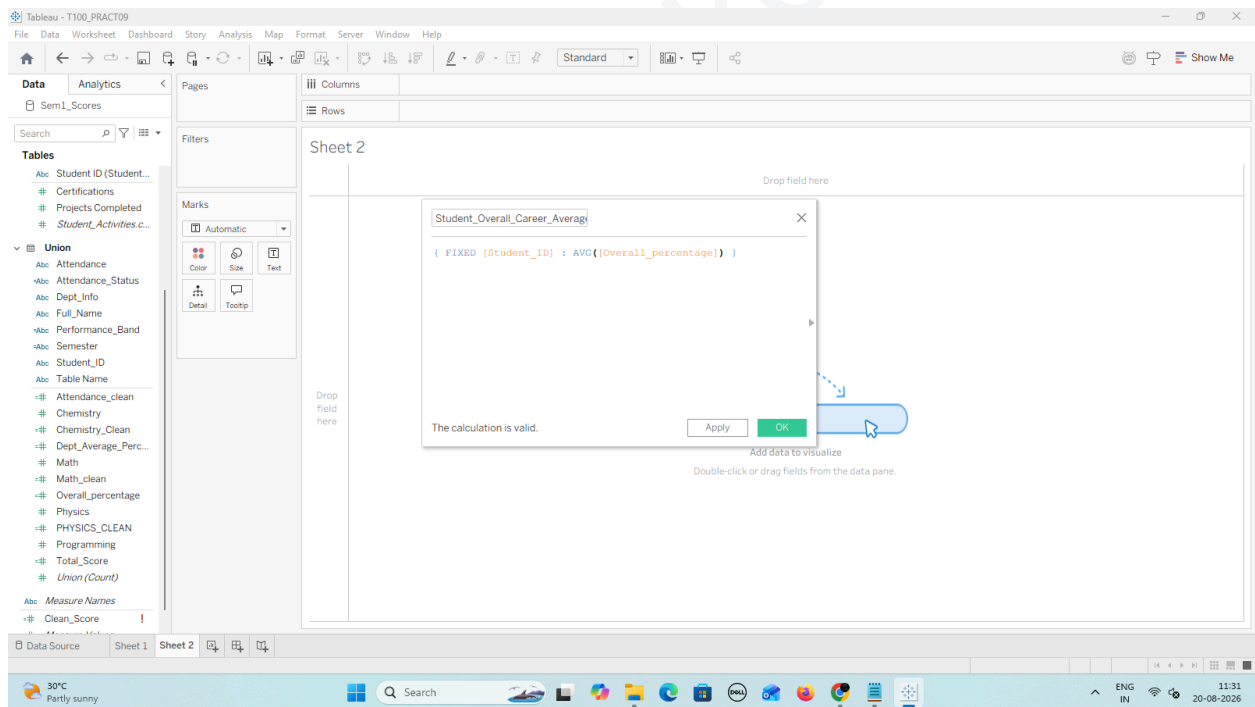
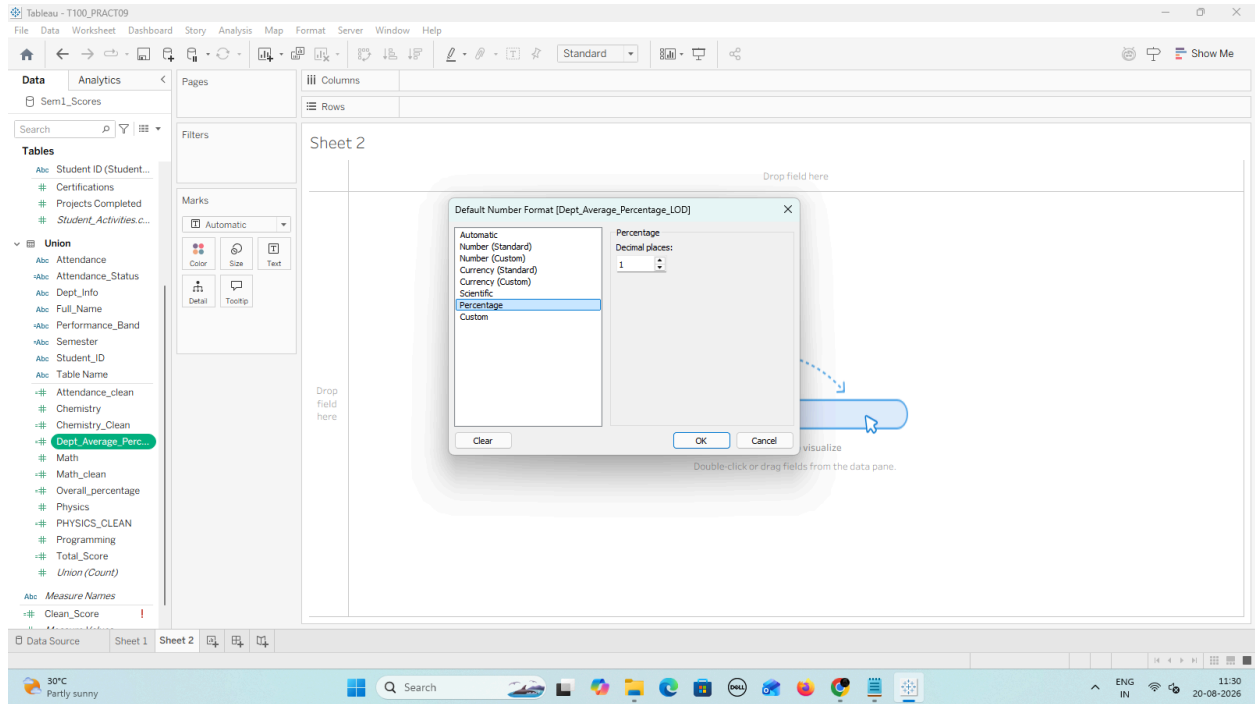
The screenshot shows the Tableau Desktop interface with a calculated table displayed. The table has columns for 'Student_ID', 'Full_Name', and 'Semester'. The data is organized into a grid with rows for 'Semester 1' and 'Semester 2'. The 'Performance_Band' is color-coded: blue for 'Needs Academic Sup...' and orange for 'Second Class'.

	S101	S102	S103	S104	S105	S106	S107	S108	S109	S110
Semester 1	Aarav Sharma 67.2%	Priya Verma 41.4%	priya VERMA 41.7%	Rohan Gupta 69.7%	Sneha Patel 38.8%	aditya Rao 66.5%	Aditya Rao 38.9%	Ananya Singh 71.5%	Rahul Nair 49.4%	Kavya Reddy 63.0%
Semester 2	69.2%	62.4%	63.2%	72.0%	43.1%	68.5%	58.7%	72.7%	51.9%	65.5%

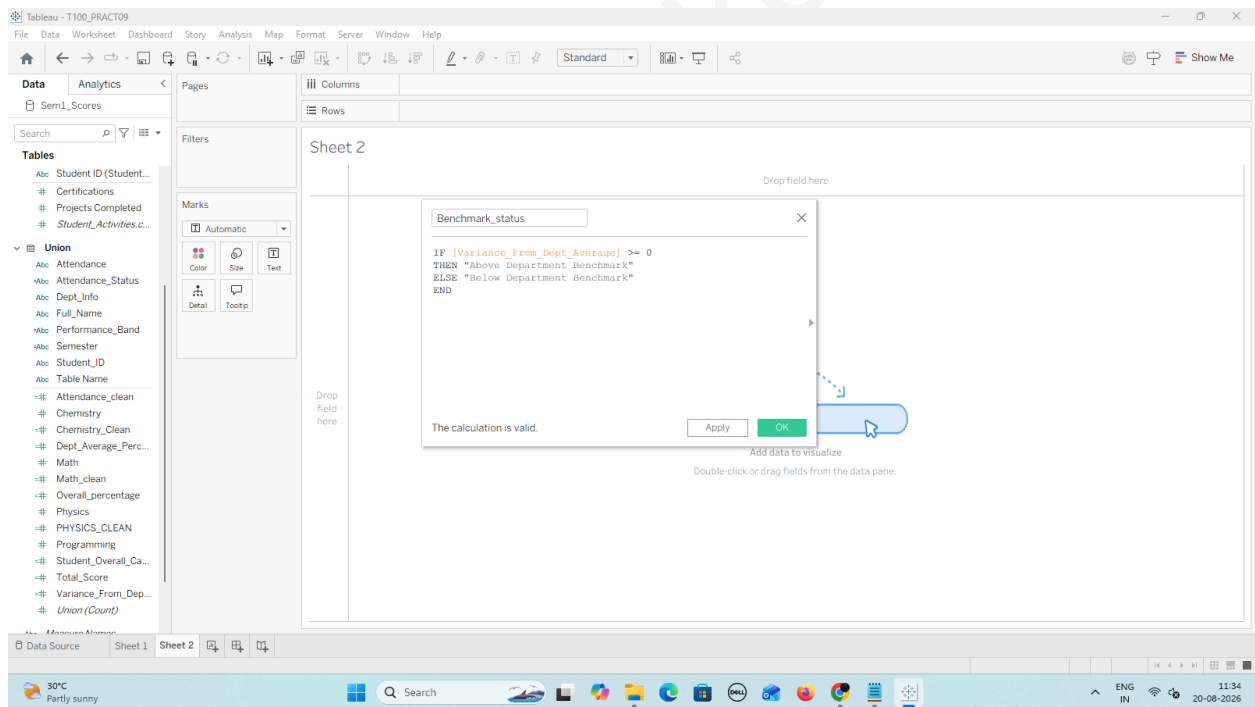
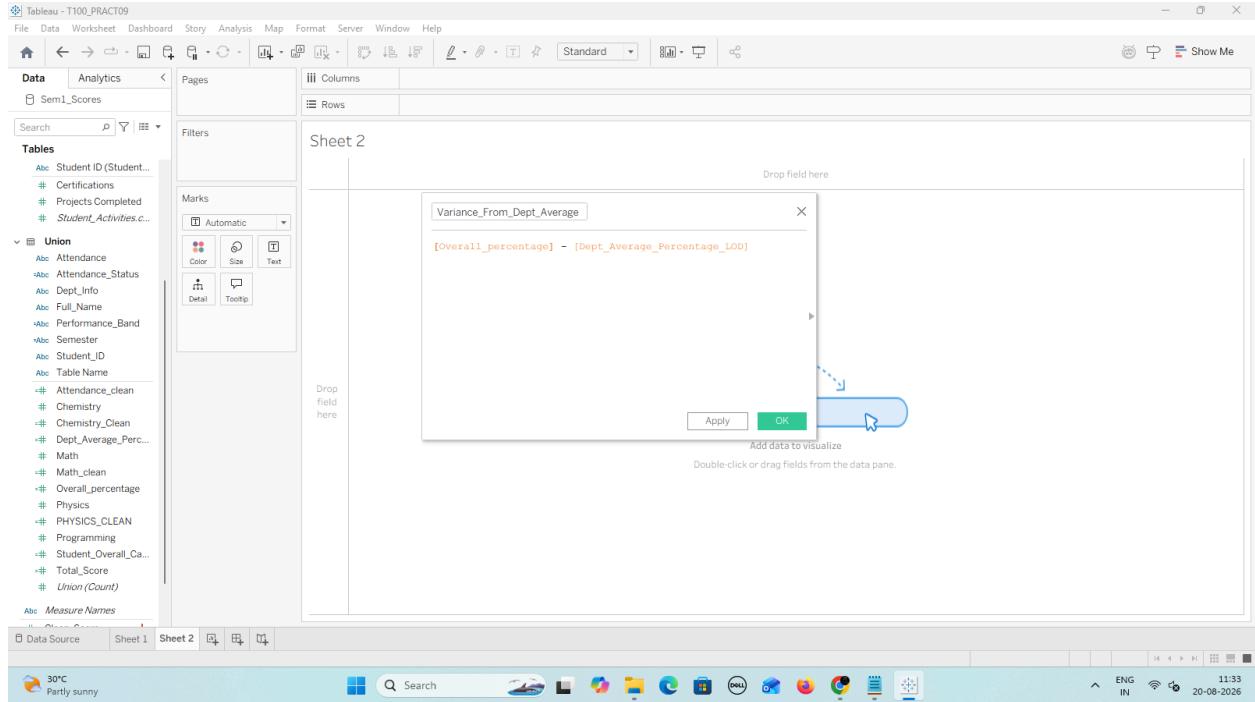
SHETH L.U.J & SIR M.V



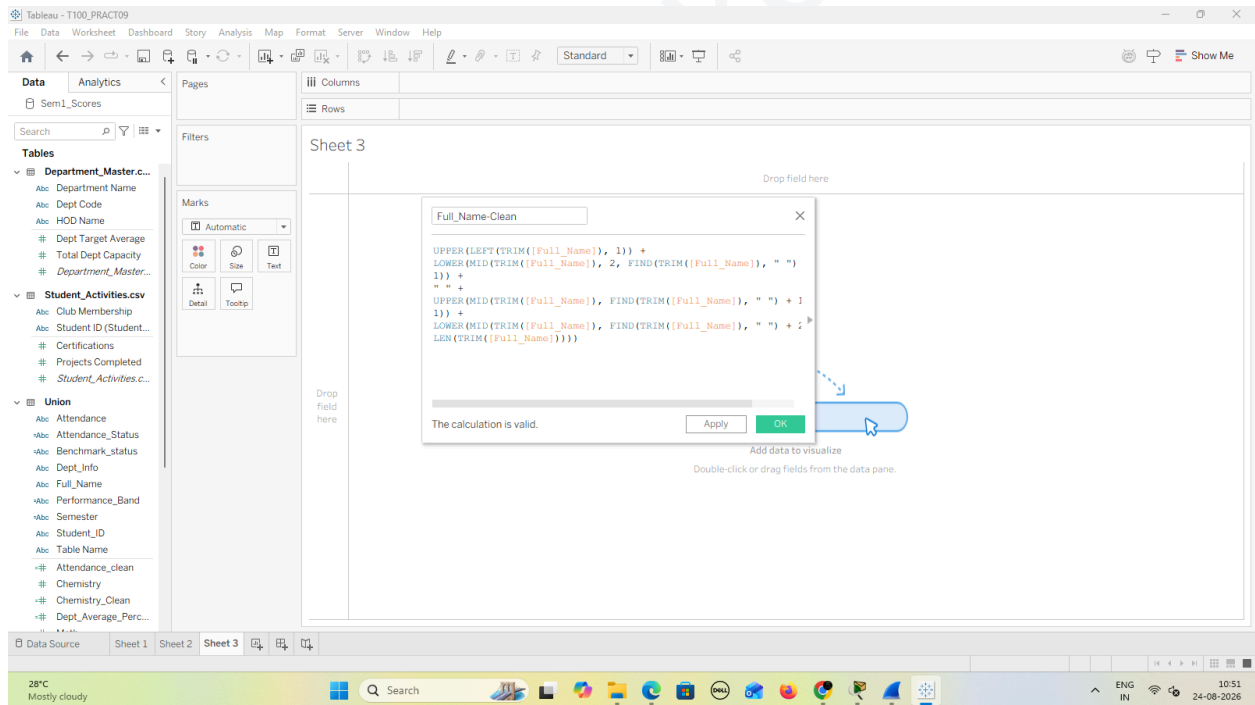
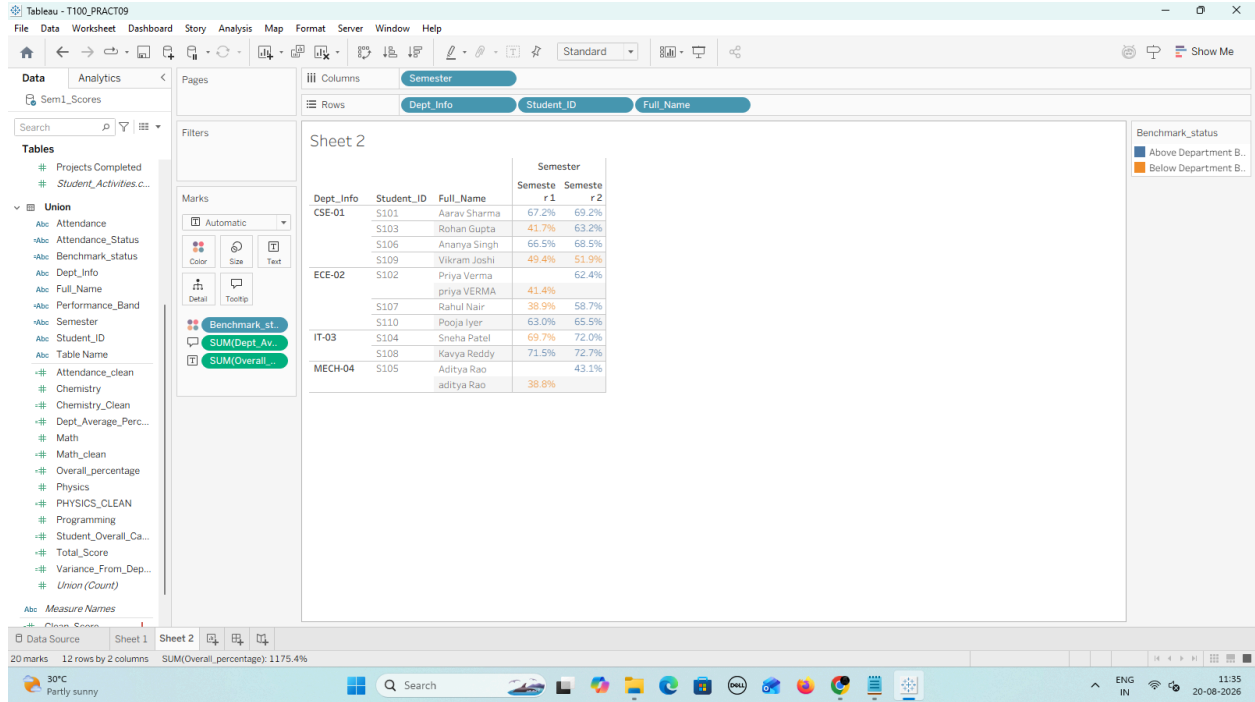
SHETH L.U.J & SIR M.V



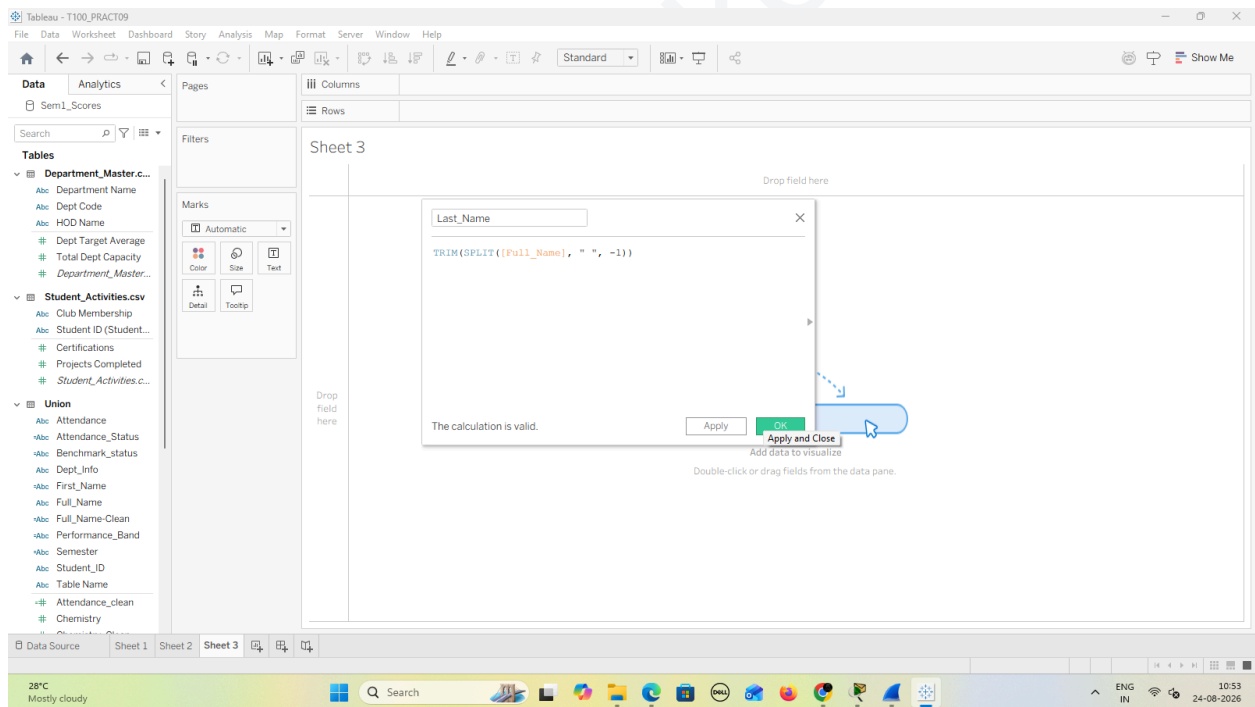
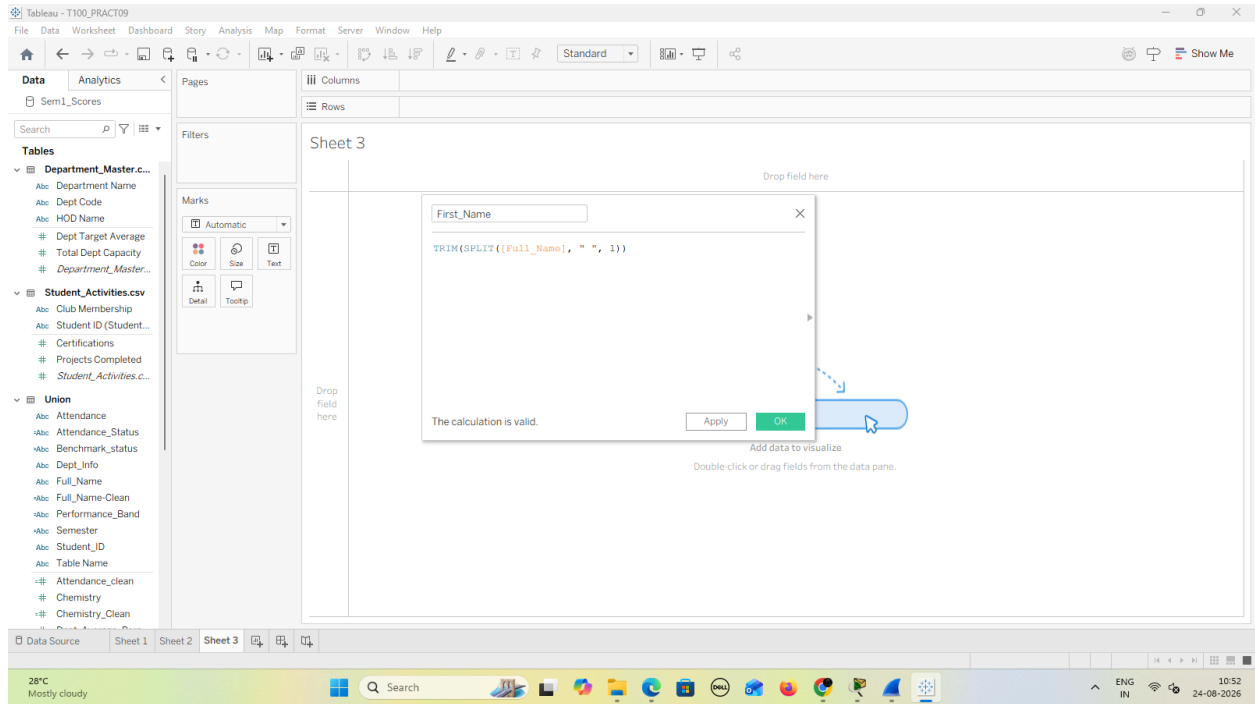
SHETH L.U.J & SIR M.V



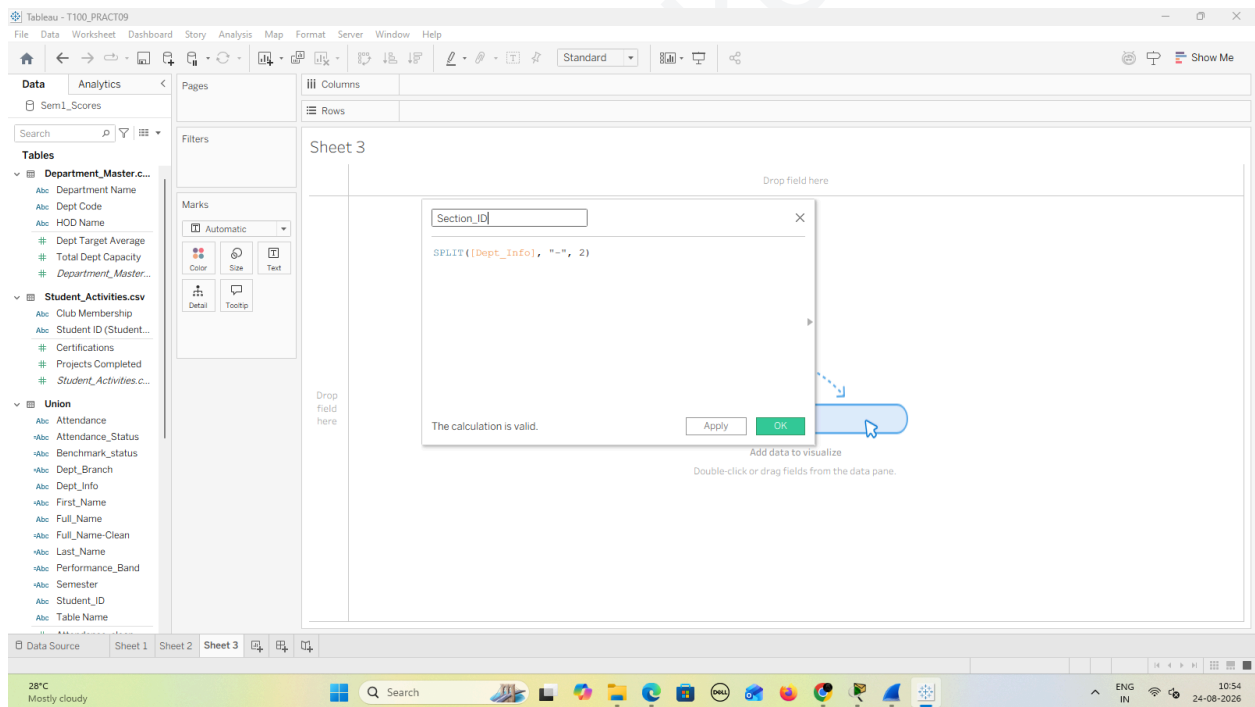
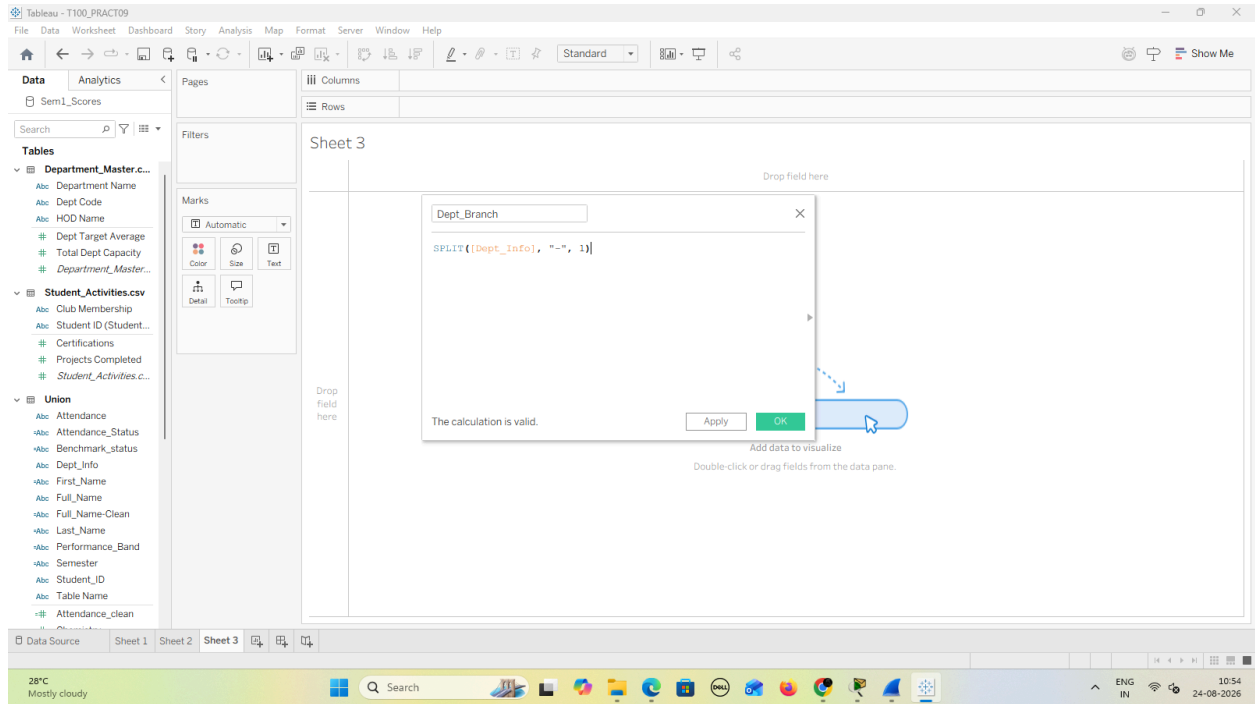
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The screenshot shows the Tableau Desktop interface with a new calculation editor open. The calculation is: `[Student_ID] + " - " + [Full_Name-Clean] + " (" + [Dept_Branc`. The interface includes a data pane on the left with tables like Department_Master and Student_Activities, a marks card, and a main view area.

The screenshot shows the final Tableau view. The Columns shelf contains 'Semester' and the Rows shelf contains 'Student_Display_Label'. The Marks card is set to 'SUM(Overall_Percentage)' with a color map for 'Performance_Band'. The view displays a table of student performance data.

Student_Display_Label	Semester	
	r 1	r 2
S102 - Priya Verma (ECE)	41.4%	62.4%
S104 - Sneha Patel (IT)	69.7%	72.0%
S105 - Aditya Rao (MECH)	38.8%	43.1%
S107 - Rahul Nair (ECE)	38.9%	58.7%
S108 - Kavya Reddy (IT)	71.5%	72.7%
S110 - Pooja Iyer (ECE)	63.0%	65.5%